Journal of the Taiwan Institute of Chemical Engineers 71 (2017) 381–387

Contents lists available at ScienceDirect

Journal of the Taiwan Institute of Chemical Engineers

journal homepage: www.elsevier.com/locate/jtice

Sustainable glucose-based phenolic resin and its curing with a DGEBA epoxy resin

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ARTICLE INFO

Article history:
Received 23 January 2016
Revised 30 October 2016
Accepted 21 November 2016

Keywords:
Phenol–hydroxymethylfurfural resin
DGEBA epoxy resin
Curing kinetics
Thermal stability
Renewable polymer
Properties

ABSTRACT

A sustainable novolac-type resin – phenol–hydroxymethylfurfural (PHMF) resin was prepared by reacting phenol with HMF, *in-situ* derived from glucose, at 120 °C by acid catalysis. Bisphenol A type epoxy resin, *i.e.* bisphenol A diglycidyl ether (DGEBA), was used as a formaldehyde-free curing agent by substituting conventional formaldehyde-based hexamethylene tetraamine (HMTA) to crosslink the PHMF resin. Curing mechanism was probed and the curing proceeded likely with the ring opening reaction between the DGEBA and reactive hydroxyl groups. DGEBA not only made this system truly formaldehyde-free but also helped form a void-free matrix which is an important merit for composites. The kinetic parameters of the curing reaction were evaluated with model-free and model-fitting methods using exothermal peak data from the curing process. The thermo-mechanical characterization of the cured resin and fiber reinforced bio-composites showed good heat resistance and mechanical performance, suggesting its potential for producing void- and formaldehyde-free composite materials.

1. Introduction

Due to increasing environmental awareness and the fast depletion of petroleum resources, the demand for the sustainable chemicals and materials (*e.g.*, bio-based plastics and polymers) derived from renewable resources is expected to increase rapidly in the next 20–30 years [1,2].

Phenol–formaldehyde resin (PF), the earliest commercial synthetic resin, has been widely used as an adhesive in engineered wood products and composites owing to its excellent mechanical properties. While most work on bio-based PF resin have been focused on phenol substitution with bio-phenols derived from lignin or lignocellulosic biomass, [3,4] formaldehyde has also provoked increasing environmental concern [5]. In fact, formaldehyde has been classified as a known human carcinogen; its permissible exposure level has been strictly regulated [6]. Due to the toxicity of formaldehyde, it is imperative to develop formaldehyde-free phenolic resins. Procuring formaldehyde-free phenolic resins can be achieved by replacing formaldehyde with green alternatives, in particular chemicals from bio-renewable resources (biomass). Glucose is abundantly available from cellulose and hemicellulose *via* hydrolysis. The authors' group has been working on the replacement of formaldehyde with HMF, *in-situ* generated from glucose *via* a catalytic process, for producing a novolac type phenol–(hydroxymethyl) furfural (PHMF) resins [7].

Conventionally, hexamethylene tetramine (HMTA) has been used as the most common compound for curing novolac type phenolic resins. However, HMTA is regarded as a hazardous air pollutant and restricted by the Federal Environmental Protection Agency of the United States in polymeric composite manufacture as it decomposes into ammonia and formaldehyde upon heating [8-10]. Thus, it is necessary to find an alternative to HMTA for novolac resins. Epoxides have a high reactivity towards a variety of chemicals or functional groups, *via* either nucleophilic or electrophilic reaction through epoxy ring opening. An example of epoxides applications is a hardener for novolac phenolic resins. Hsieh and Su studied cure kinetics of epoxy–novolac molding while forming secondary alcohols by proton transfer [11]. Researchers further confirmed that the secondary alcohols formed could easily react with the epoxy groups [12,13]. Han and coworkers reported the curing of phenol novolac with biphenyl epoxy resin at an equivalent mass [14].] Polyphenol-generated epoxy resins with a high glass transition temperature are frequently used as a hardener [15]. Gagnebien et al. used bisphenol A and glycidylethers as models of epoxy and phenol, respectively, to study the hydroxyetherification reactions, where the formation of the hydroxy branch was confirmed [16], as similarly reported by Alevey [17] and Burchard et al. [18].

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http://dx.doi.org/10.1016/j.jtice.2016.11.025